A six-variable cyclic quartic inequality with a variance remainder

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Abstract

For every real six-tuple, in its specified cyclic order, we prove

$$\left(\sum_{i=1}^{6} x_i^2\right)^2 - \left(\sum_{i=1}^{6} x_i\right)\left(\sum_{i=1}^{6} x_i x_{i+1}^2\right) \geq \frac{1}{4}\left(\sum_{i=1}^{6}(x_i - \bar{x})^2\right)^2, \qquad x_7 = x_1.$$

Here $\bar{x}$ is the arithmetic mean. In particular, this resolves the six-variable $(a, b) = (1, 2)$ case discussed numerically by Javaheri and classifies equality: all six entries must be equal. The proof uses a decomposition into opposite pairs, Cauchy–Schwarz, and an elementary eigenvalue bound for a trace-zero symmetric 3×3 matrix. It does not require numerical optimization or a computer certificate. A separately verified rational sum-of-squares certificate is included as a supplementary check of the original inequality.

1 Statement and scope

For $x = (x_1, \ldots, x_6) \in \mathbb{R}^6$, with subscripts interpreted modulo 6, put

$$S(x) = \sum_{i=1}^{6} x_i, \qquad Q(x) = \sum_{i=1}^{6} x_i^2, \qquad C(x) = \sum_{i=1}^{6} x_i x_{i+1}^2, \qquad F(x) = Q(x)^2 - S(x)C(x).$$

No ordering or permutation of the entries is allowed or needed.

Theorem 1. *For every $x \in \mathbb{R}^6$, writing $t = S(x)/6$, one has*

$$F(x) \geq \frac{1}{4}\left(Q(x) - S(x)^2/6\right)^2 = \frac{1}{4}\left(\sum_{i=1}^{6}(x_i - t)^2\right)^2. \tag{1}$$

Consequently $S(x)C(x) \leq Q(x)^2$, with equality if and only if $x_1 = \cdots = x_6$.

The original nonnegative domain $[0, \infty)^6$ is a subset of the theorem's domain. Thus zero coordinates, every boundary face, and the all-zero vector are covered directly. The constant $1/4$ in the remainder is a convenient explicit constant; no optimality claim is made for it.

Javaheri studies a larger parameterized cyclic inequality in [1]. Its Section 8 reports numerical evidence for $(a, b, n) = (1, 2, 6)$, without a global proof of that case. The necessary second-variation condition in Theorem 1.2 is degenerate at these parameters. Theorem 1 proves this case and extends its domain to real entries. It does not resolve the full parameter conjecture. The earlier Javaheri–Shen result [2] concerns $(a, b) = (1, 1)$ and has different parameters.

An audit on 28 October 2026 of the cited versions and related search results did not identify a prior proof of this exact case. This records the scope of the search and does not establish universal priority.

2 Centering the quartic

Set

$$u_i = x_i - t, \qquad \sum_{i=1}^{6} u_i = 0, \qquad q = \sum_{i=1}^{6} u_i^2, \quad r = \sum_{i=1}^{6} u_i u_{i+1}, \quad c = \sum_{i=1}^{6} u_i u_{i+1}^2.$$

Expansion of the three factors gives

$$S(x) = 6t, \qquad Q(x) = 6t^2 + q,$$
$$C(x) = 6t^3 + t(q + 2r) + c.$$

For the last identity, the quadratic terms in t vanish because $\sum u_i = 0$; the terms linear in t are $t\sum(u_{i+1}^2 + 2u_i u_{i+1})$. Therefore

$$F(x) = q^2 - 6tc + 6t^2 D, \qquad D = q - 2r. \tag{2}$$

We will control c and D using the same component of u.

3 Opposite pairs and the cubic estimate

Decompose $u = e + o$ by

$$e_i = \frac{u_i + u_{i+3}}{2}, \qquad o_i = \frac{u_i - u_{i+3}}{2}.$$

There are real numbers a, b, h, p, v, w such that

$$e = (a, b, h, a, b, h), \qquad o = (p, v, w, -p, -v, -w), \qquad a + b + h = 0. \tag{3}$$

The two vectors are orthogonal. Define

$$s = \|e\| = \sqrt{2(a^2 + b^2 + h^2)}.$$

In particular,

$$q = s^2 + 2(p^2 + v^2 + w^2) = s^2 + \|o\|^2. \tag{4}$$

Lemma 2. *For the decomposition above,*

$$D = 2s^2 + 2(p - v + w)^2 \geq 2s^2. \tag{5}$$

Proof. Mixed adjacent products cancel in pairs, so

$$r = 2(ab + bh + ha) + 2(pv + vw - wp).$$

Since $a + b + h = 0$, we have $2(ab + bh + ha) = -(a^2 + b^2 + h^2)$. Substituting this and (4) into $D = q - 2r$ gives

$$D = 4(a^2 + b^2 + h^2) + 2(p^2 + v^2 + w^2 - 2pv - 2vw + 2wp),$$

which is (5). □

Lemma 3. *If M is a real symmetric 3×3 matrix with $\operatorname{tr} M = 0$, then*

$$\|M\|_{\mathrm{op}}^2 \leq \frac{2}{3} \operatorname{tr}(M^2).$$

Here $\|M\|_{\mathrm{op}}$ is the largest absolute value of an eigenvalue.

Proof. Its eigenvalues are real. For any one of them, denoted λ, write the other two as μ, ν. Their sum is $-\lambda$, and hence

$$\mu^2 + \nu^2 \geq \frac{(\mu+\nu)^2}{2} = \frac{\lambda^2}{2}.$$

Thus $\operatorname{tr}(M^2) = \lambda^2 + \mu^2 + \nu^2 \geq 3\lambda^2/2$. Apply this to each eigenvalue. □

Lemma 4. *The centered cubic satisfies $|c| \leq sq$.*

Proof. Under a shift of three indices, e is unchanged and o changes sign. Consequently all terms containing an odd number of factors from o cancel in the expansion of $c = C(e+o)$. The remaining terms are

$$\begin{aligned} c &= C(e) + \sum_{i=1}^{6} e_i o_{i+1}^2 + 2\sum_{i=1}^{6} o_i e_{i+1} o_{i+1} \\ &= C(e) + 2(av^2 + bw^2 + hp^2) + 4(bpv + hvw - apw) \\ &= C(e) + 2z^{\mathsf{T}} M z, \end{aligned} \tag{6}$$

where

$$z = \begin{pmatrix} p \\ v \\ w \end{pmatrix}, \qquad M = \begin{pmatrix} h & b & -a \\ b & a & h \\ -a & h & b \end{pmatrix}.$$

The orientation of the cycle is preserved in (6). Since $a + b + h = 0$,

$$\operatorname{tr} M = 0, \qquad \operatorname{tr}(M^2) = 3(a^2 + b^2 + h^2) = \tfrac{3}{2}s^2.$$

Lemma 3 gives $\|M\|_{\mathrm{op}} \leq s$, so

$$|2z^{\mathsf{T}} M z| \leq 2s \|z\|^2 = s \|o\|^2 . \tag{7}$$

For the purely even part, Cauchy–Schwarz gives

$$|C(e)| \leq \left(\sum_{i=1}^{6} e_i^2\right)^{1/2} \left(\sum_{i=1}^{6} e_{i+1}^4\right)^{1/2} \leq s^3. \tag{8}$$

The second inequality follows from $\sum e_i^4 \leq (\sum e_i^2)^2$. Combining (6)–(8) with (4) yields $|c| \leq s^3 + s\|o\|^2 = sq$, as required. □

4 Proof of the theorem and examples

Proof of Theorem 1. Using (2) and Lemmas 2 and 4, we obtain, for either sign of t,

$$\begin{aligned} F(x) &= q^2 - 6tc + 6t^2 D \\ &\geq q^2 - 6|t|sq + 12t^2 s^2 \\ &= 12\left(|t|s - \frac{q}{4}\right)^2 + \frac{q^2}{4} \geq \frac{q^2}{4}. \end{aligned}$$

Finally $q = Q(x) - S(x)^2/6$ by centering, proving (1). If $F(x) = 0$, the bound forces $q = 0$, so all entries equal t. Conversely a constant tuple has $Q(x)^2 = (6t^2)^2 = (6t)(6t^3) = S(x)C(x)$. □

For a nonnegative nonzero vector one may divide by $Q(x)^2$ to get

$$\frac{S(x)C(x)}{Q(x)^2} \le 1 - \frac{1}{4}\left(1 - \frac{S(x)^2}{6Q(x)}\right)^2.$$

The division is unnecessary in the proof and excludes no point from the theorem. In particular:

- If $x = (t, t, t, t, t, t)$, then $q = 0$ and $F(x) = 0$, including $t = 0$.
- If $x = (1, 0, 0, 0, 0, 0)$, then $C(x) = 0$, $F(x) = 1$, and $q = 5/6$. The remainder bound gives $F(x) \ge 25/144$.
- If $S(x) = 0$, then $F(x) = Q(x)^2 = q^2$, so every nonzero such vector satisfies the original inequality strictly.

5 Independent checks and reproducibility

The proof above is analytic and self-contained. Its essential steps are the centering identity, the opposite-pair identities, the trace-zero matrix bound, and completion of a square. All divisions used to define coordinates are by fixed positive integers; no positivity assumption on the entries or nonzero-coordinate assumption occurs.

Two independent algebraic checks supplement this proof. First, a coefficient verifier expands the centering and pair identities using exact rational polynomial arithmetic. The substitution

$$x = (t + a + p,\ t + b + v,\ t - a - b + w,\ t + a - p,\ t + b - v,\ t - a - b - w)$$

parametrizes every real six-tuple, so these checks concern unrestricted identities. Second, a separate exact Gram certificate proves $F \ge 0$. Let $z(x)$ be the vector of the 21 quadratic monomials $x_i x_j$ in lexicographic order $1 \le i \le j \le 6$. The supplementary data give rational matrices $B \in \mathbb{Q}^{21\times 18}$ and $H \in \mathbb{Q}^{18\times 18}$ such that

$$F(x) = z(x)^{\mathsf{T}} B H B^{\mathsf{T}} z(x), \qquad H = L\Delta L^{\mathsf{T}}, \qquad \Delta_{jj} > 0.$$

Here L is unit lower triangular and Δ is diagonal. The verifier checks all 126 degree-four monomial coefficients, the matrix factorization, and the signs of its 18 rational pivots. Thus

$$F(x) = \sum_{j=1}^{18} \Delta_{jj}\big((L^{\mathsf{T}} B^{\mathsf{T}} z(x))_j\big)^2.$$

This certificate verifies the original inequality independently; the variance remainder is proved by the analytic argument above. No floating-point solver output is accepted as an exact certificate.

From the supplement's root directory, the following commands require only Python's standard library:

```
python src/verify_identities.py
python src/verify_certificate.py
```

The latter reads `certificates/exact_gram.json`; all paths are resolved relative to the package, not to the current working directory. To rebuild this document, run the supplied build script or compile `paper/main.tex` twice with pdfLaTeX. The source, final PDF, verification reports, independent Fourier derivation, and literature audit are included in the package. The main proof is complete without the supplementary certificate or any software.

Verification status. This note was developed by GPT-6 Astra, an AI system, including parallel derivations and exact-arithmetic checks. Those checks constitute internal verification. No external peer review or proof-assistant formalization is claimed.

References

[1] M. Javaheri, *A new arrangement inequality (revised and extended version)*, arXiv:2610.33653v1, 2026, especially Theorem 1.2 and Section 8. `https://arxiv.org/abs/2610.33653v1`.

[2] M. Javaheri and H. Shen, *On a Cyclic Inequality Related to Chebyshev Polynomials*, arXiv:2301.00679v1, submitted 2022, especially Theorem 2. `https://arxiv.org/abs/2301.00679v1`.